

Kevin Caldwell

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Education

University of Toronto, Toronto, Ontario

Sept. 2022 to April 2027

Bachelor of Applied Science in Engineering Science

Major: **Electrical and Computer Engineering** (PEY)

GPA: 3.45

Skills

Programming Languages & HDL

C/C++, Python, RISC-V and NIOS V ASM, ARM C, CUDA, System Verilog, Verilog

Software and Platforms

STM32, Git, GitHub, Quartus Prime, Model Sim/Questa, Altera MAX10 and Cyclone V FPGA

Professional Experience

Software Developer

Primate Labs Inc. | Toronto, Ontario

(May. 2025 to August 2026)

- Researched several libraries and utilities for running LLM's locally on MacOS and Linux integrate into Geekbench for the Gen AI benchmark.
- Designed a Generative AI framework for accommodating various backends and libraries such as MLX, OpenVINO, LiteRT/TFLite, Llama.cpp, Windows ML and CoreAI to enable cross-platform support for the Gen AI benchmark.
- Wrote TFLite CPU and GPU layers for running the audio transcription model Whisper, updating the Speech Recognition workload to a transformer model from an NLP model to benchmark speech recognition for CPU.
- Added support for RISC-V vector instructions in the [Jolt Physics library](#), to enable SIMD acceleration in RISC-V targets for the Game Physics workload.
- Researched LiteRT and Windows ML libraries to integrate into Geekbench to modernize Inference libraries for AI workloads.

Researcher

Dynamic Graphics Project | Toronto, Ontario

(Oct. 2024 to Present)

- Implemented Advanced Algorithms like **Genetic Algorithm**, **TSP heuristics** using Python to model Hybrid Truck-Drone Delivery Systems (HTDDS) energy consumption.
- Modelled Dynamic System Simulations to study the development of more efficient HTDDS.
- Enhanced Program Efficiency using **CUDA C/C++**, resulting in a 10 times reduction in execution time.

Edward S. Rogers Sr. Department (Professor Mark Jeffrey) | Toronto, Ontario

(June 2025 to Present)

- Implemented several single threaded and multithreaded variations of Maximal Independent Set (MIS) to run on the PBBS
- Benchmarked various solutions to the MIS problem on the PBBS benchmark, comparing them to the proposed countingMIS solution
- Profiled multiple single threaded and multi-threaded MIS solutions using the perf tool.

University of Toronto Aerospace Team (UTAT) Space Systems Division | Toronto, Ontario

Software Developer

(May 2024 to Present)

- Successfully prototyped an SD Card file system configuration using FatFS, DMA, and SDMMC modules, optimizing hyperspectral image storage for maximum efficiency and data integrity. Achieved impressive read/write speeds of up to 95% of the theoretical limit.
- Implemented CCSDS 123.0-B-2 compression algorithm using **Embedded C**, significantly reducing data storage requirements for hyperspectral images. Achieved 50% compression on both synthetic and non-synthetic test data

Electrical and Computer Engineering University of Toronto | Toronto, Ontario

Student

(Oct. 2022 to Present)

- Implemented a Single Bus Processor using **System Verilog** for the DE10-Lite to implement the **RV32I Compute Instructions**, achieving 3 clock cycles per Instruction, as an extension to the ECE253: Digital Systems and Computer Organization Course.
 - Wrote Firmware in **NIOS V Assembly and C** to program various peripherals, including Switches and 7-Segment Displays, JTAG UART, Timer and VGA Peripherals to create simple Interactive Programs, as part of ECE352: Computer Organization.
 - Designed a **Vector Extension** for a custom 8-bit RISC ISA, including Load, Compute and Store operations in Verilog for a project in ECE352: Computer Organization
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Publications

- Beyond the Visible: Jointly Attending to Spectral and Spatial Dimensions with HSI-Diffusion for the FINCH Spacecraft
- Balancing Trade-Offs: The Energy Efficiency of Unmanned Aircraft Systems Integration in Last-Mile Delivery and Operational Policy Restrictions

Awards

- Don Walker Magna Scholarship
- Honor Roll: Cayley Contest 2020, Fryer 2019, CIMC 2019

Skills (cont.)

Programming Languages & HDL Swift, MATLAB,
Software and Platforms Windows, Linux, MacOS, MS Office

Professional Experience (cont.)

Software Developer

Primate Labs Inc. | Toronto, Ontario *(May. 2025 to August 2026)*

- Ran LLM’s locally on Apple devices using Apple’s MLX framework in Python to prototype code related to the MLX backend for the Geekbench Gen AI Benchmark.
- Integrated Intel OpenVINO generative AI library into Geekbench Gen AI, resulting in up to 10x performance on x86 systems.
- Added unit tests to test GPU kernels for various GPU workloads, which helped introduce parity between CPU and GPU, improving accuracy scores across several GPU workloads such as Fluid Simulation, Horizon Detection and Particle Physics.
- Implemented Text Summarization, Text Improvement, Reasoning, Code Generation and Code Repair workloads to Gen AI benchmark to introduce workloads that reflect common tasks performed using LLMs.

Researcher

Dynamic Graphics Project | Toronto, Ontario *(Oct. 2024 to Present)*

- Implemented **Socket IPC** for efficient thread collaboration, boosting simulation performance.
- Enhanced benchmarking capabilities by creating diverse synthetic test environments using Voronoi Maps and Linear Masks.

University of Toronto Aerospace Team (UTAT) Space Systems Division | Toronto, Ontario

Software Developer *(May 2024 to Present)*

- Efficiently programmed **STM32H7 DCMI and DMA** peripherals to develop image acquisition firmware for the FINCH Nanosatellite payload camera, ensuring lossless pixel capture

Systems Engineer

- Performed **FMEA** for developing robust FDIR (Fault Detection, Isolation, Recovery) to ensure satellite operations during critical events.

Robotics for Space Exploration at UofT | Toronto, Ontario

Science Team Member *(June 2025 to Present)*

- Used CAN communication to transmit data between Arduinos and an STM32G4 to prototype CAN communication for the Rover.
 - Designed a CAN library to enable complex communication between various subsystems and their peripherals.
 - Designed an extension to the CAN library for sending contiguous data blocks up to size 65 kB from Arduino Nano to a Raspberry Pi for sending spectrometer pixel data to meet the science objective of transmitting soil spectrometer data.
 - Wrote code for various peripherals connected to modules to meet functional requirements for Science mission.
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